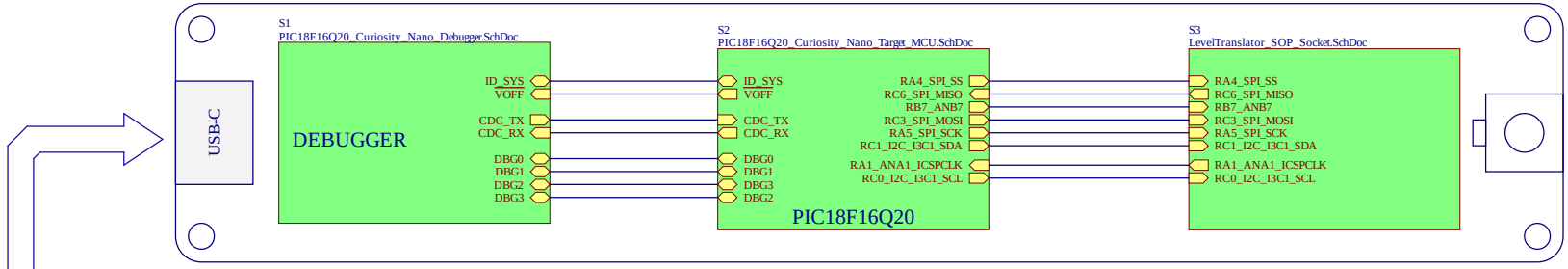
PIC18F16Q20 Curiosity Nano



On-Board Peripherals		
LED0	RC7	Active Low
SW0	RA2	Active Low

Program / Debug Interface	
ICSP DAT	RA0
ICSP CLK	RA1
MCLR	RA3

IDE
Program Debug

Mass Storage
CURIOSITY (E:)

Drag & Drop Programming
helloworld.hex

Serial / CDC port	
UART[n]* TX	RC4
UART[n]* RX	RC5

*configure with Peripheral Pin Select (PPS)

Terminal
> Hello World!

DGI (Data Gateway Interface)	
Debug GPIO 0	RA2 (SW0)

Data Visualizer

TEST FIXTURE1



Curiosity Nano Test Fixture

Test1



TEST, PIC18F16Q20-CNANO

Test2



FW, CNANO-DEBUGGER

LABEL1



[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

PCBA LABEL 18X6mm

"Text in Silkscreen on PCB"

CNANO Template Revision: 2.3

Project Owner: ST	
PCB Layout Contact: TF	
PartNumber: EV73T25A	Project Title: *PIC18F16Q20 Curiosity Nano* — Variant: [No Variations]
Sheet Title: Top Level	
Size: A3 — SCH #: 02-00457 — Rev: 3 — Date: 2026-10-30	
PCB #: 04-11563 — Rev: 3 — Sheet 1 of 5	
File: PIC18F16Q20_Curiosity_Nano_TopLevel.SchDoc	

Designed with Altium — Altium.com

S3
PIC18F16Q20_Curiosity_Nano_Revision_History.SchDoc

PIC18F16Q20

	PIC18F16Q20	
Debugger	*Name*	*Pin*
CDC TX	UART[n] RX	**RC5**
CDC RX	UART[n] TX	**RC4**
DBG0	ICSPDAT	**RA0**
DBG1	ICSPCLK	**RA1**
DBG2	SW0 / GPIO0	**RA2**
DBG3	MCLR	**RA3**
VTG	**1.8V - 3.6V**	

RA4_SPI_SS
RC6_SPI_MISO
RB7_ANB7
RC3_SPI_MOSI
RA5_SPI_SCK
RC1_I2C_I3C1_SDA
RA1_ANA1_ICSPCLK
RC0_I2C_I3C1_SCL

PIC18F16Q20

U100
PIC18F16Q20

RA3_MCLR
RC5_UART_RX
RC4_UART_TX
RC3_SPI_MISO
RC6_SPI_MISO
RC7_ANC7_LED0_PWM
RB7_ANB7
RB6_I3C2_SCL
RB5_I3C2_SDA

1 RA3/VPP/MCLR
2 RC5
3 RC4
4 RC3
5 RC6
6 RC7
7 RB7
8 RB6
9 RB5
10 VDDIO3
EPAD 21
20 OSC2/RA4
19 OSC1/RA5
18 VDD
17 VSS
16 ICSPDAT/RA0
15 ICSPCLK/RA1
14 RA2
13 RC0
12 RC1
11 VDDIO2

RA4_SPI_SS
RA5_SPI_SCK
RA0_ANA0_ICSPDAT
RA1_ANA1_ICSPCLK
RA2_ANA2_DBG2_SW0_INT0
RC0_I2C_I3C1_SCL
RC1_I2C_I3C1_SDA

VCC_TARGET
C100 0.1uF
GND

VCC_VDDIO3_P1V2
R105 0R
VDDIO3
J107
C101 0.1uF

VCC_TARGET
R106 0R
VDDIO2
J108
C102 0.1uF

J208 can be used to apply external power to VDDIO3, and J207 can be used to apply external power to VDDIO2.
Make sure to remove R200/R206 to disconnect on-board power before applying power to J208/207.

VDDIO2 powers MVIO pins RC0 and RC1

VDDIO3 powers MVIO pins RB5 and RB6

DEBUGGER CONNECTIONS

CDC_RX
CDC_TX
DBG0
DBG3
DBG1
DBG2
VOFF
ID_SYS

PROG/DEBUG Pull

DBG0
DBG1
R100 47k
R101 47k
GND

MCLR Pull

VCC_TARGET
R102 47k
RA3_MCLR

J100
DEBUGGER
1 RESERVED
2 ID
3 CDC RX
4 CDC TX
5 DBG1
6 DBG2
7 0 TX
8 1 RX
9 2 SDA
10 3 SCL
11 4 MOSI
12 5 MISO
13 6 SCK
14 7 SS
15 GND
30 VBUS
29 VOFF
28 DBG3
27 DBG0
26 GND
25 VCC
24 ADC 7
23 ADC 6
22 ADC 5
21 PWM 4
20 PWM 3
19 ADC2
18 ADC1
17 ADC0
16 GND
TARGET
EDGE 30P Female CNANO

VCC_VBUS
VCC_EDGE
J101 J103 J105 J106
J102 J104

RC4_UART_TX
RC5_UART_RX
RA1_ANA1_ICSPCLK
RA2_ANA2_DBG2_SW0_INT0
RC4_UART_TX
RC5_UART_RX
RC1_I2C_I3C1_SDA
RC0_I2C_I3C1_SCL
RC3_SPI_MOSI
RC6_SPI_MISO
RA5_SPI_SCK
RA4_SPI_SS

RA0_ANA0_ICSPDAT
RA1_ANA1_ICSPCLK
RB5_I3C2_SDA
RB6_I3C2_SCL
RC7_ANC7_LED0_PWM
RB7_ANB7
RA2_ANA2_DBG2_SW0_INT0

NOTE on I2C:

No pull-ups on board. Pull-ups should be mounted close to client device(s).

NOTE on UART/CDC:

RX/TX on the header denotes the input/output direction of the signal respective to it's source.

CDC TX is output from the DEBUGGER.
CDC RX is input to the DEBUGGER.
TX is output from the TARGET device.
RX is input to the TARGET device.

RA2_ANA2_DBG2_SW0_INT0
RC7_ANC7_LED0_PWM

USER BUTTON

R103 1k
SW100
TACT SPST
GND
"SW0"
"RA2"

USER LED

VCC_TARGET
D100
Yellow LED
R104 1k
"LED0"
"RC7"

TARGET BULK

VCC_EDGE
C103 2.2uF
GND

VDDIO3 REGULATOR

VCC_TARGET
U101
MCP1711/1.2V
VCC_VDDIO3_P1V2
C104 1uF
VIN
VOUT
EP
SHDN
GND
C105 1uF
"1.2V"

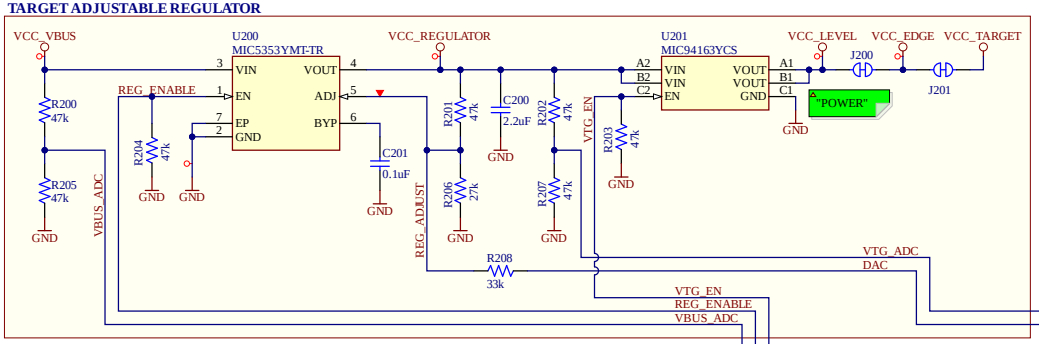
CNANO Template Revision: 2.3

Project Owner:	ST
PCB Layout Contact:	TF
PartNumber:	EV73T25A
Project Title	*PIC18F16Q20 Curiosity Nano*
Variant:	[No Variations]
Sheet Title	Target MCU
Size	A3
SCH #:	02-00457
PCB #:	04-11563
Rev:	3
Rev:	3
Date:	2026-10-30
Sheet	2 of 5
File:	PIC18F16Q20_Curiosity_Nano_Target_MCU.SchDoc

Designed with Altium
Altium.com

DEBUGGER

TARGET ADJUSTABLE REGULATOR



J200:
- Cut-strap used for full separation of target power from the level shifters and on-board regulators.
- For current measurements using an external power supply, this strap could be cut for more accurate measurements. Leakage back through the switch is in the micro ampere range.

J201:
- For current measurements using the on-board power supply, this strap must be cut and an ammeter connected across.

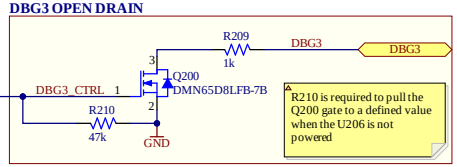
MIC5353:
Vin: 2.6V to 6V
Vout: 1.25V to 5.1V
Imax: 500mA
Dropout (typical): 50mV@150mA, 160mV @ 500mA
Accuracy: 2% initial
Thermal shutdown and current limit

Maximum output voltage is limited by the input voltage and the dropout voltage in the regulator.

Signal \ Interface	ICSP TARGET	UPDI TARGET	SWD TARGET
CDC TX	UART RX	UART RX	UART RX
CDC RX	UART TX	UART TX	UART TX
DBG0	DAT	UPDI	SWDAT
DBG1	CLK	GPIO	SWCLK
DBG2	GPIO	GPIO	SWO/GPIO
DBG3	MCLR	RESET	RESET
VCC	-	-	-

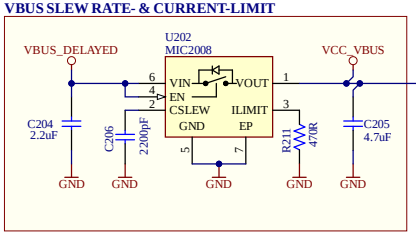
ADJUSTABLE OUTPUT AND LIMITATIONS:
- The DEBUGGER can adjust the output voltage of the regulator between 1.25V and 5.1V to the target.
- The voltage output is limited by the input (USB), which can vary between 4.40V to 5.25V
- The level shifters have a minimal voltage level of 1.65V and will limit the minimum operating voltage allowed for the target to still allow communication.
- The MIC94163 has a minimal volatege level of 1.70V and will limit the minimum voltage delivered to the target.
- Firmware configuration will limit the voltage range to be within the the target specification.

DBG3 OPEN DRAIN



R210 is required to pull the Q200 gate to a defined value when the U206 is not powered

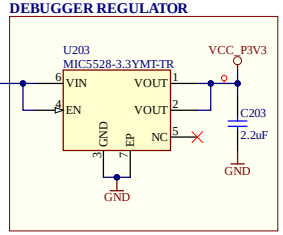
VBUS SLEW RATE- & CURRENT-LIMIT



SLEW RATE LIMIT:
- With C206 set to 2200pF, the slew rate of VCC_VBUS is limited to 2 V/ms by the power switch MIC2008.

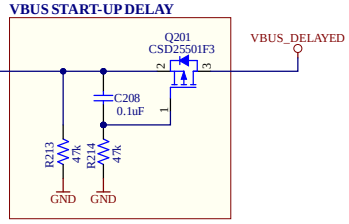
CURRENT LIMIT:
- With R211 set to 470ohm, the current through the power switch MIC2008 is limited to 500mA.

DEBUGGER REGULATOR



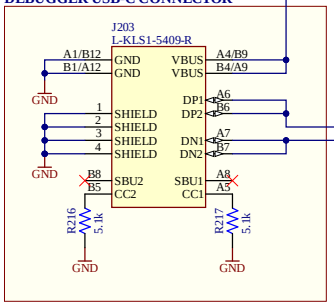
MIC5528:
Vin: 2.5V to 5.5V
Vout: Fixed 3.3V
Imax: 500mA
Dropout: 260mV @ 500mA

VBUS START-UP DELAY



START-UP DELAY::
- With C208 set to 0.1uF and R214 set to 47kohm, the P-MOS is delaying the turn-on of VCC_DELAYED by ~1ms, compared to VBUS.

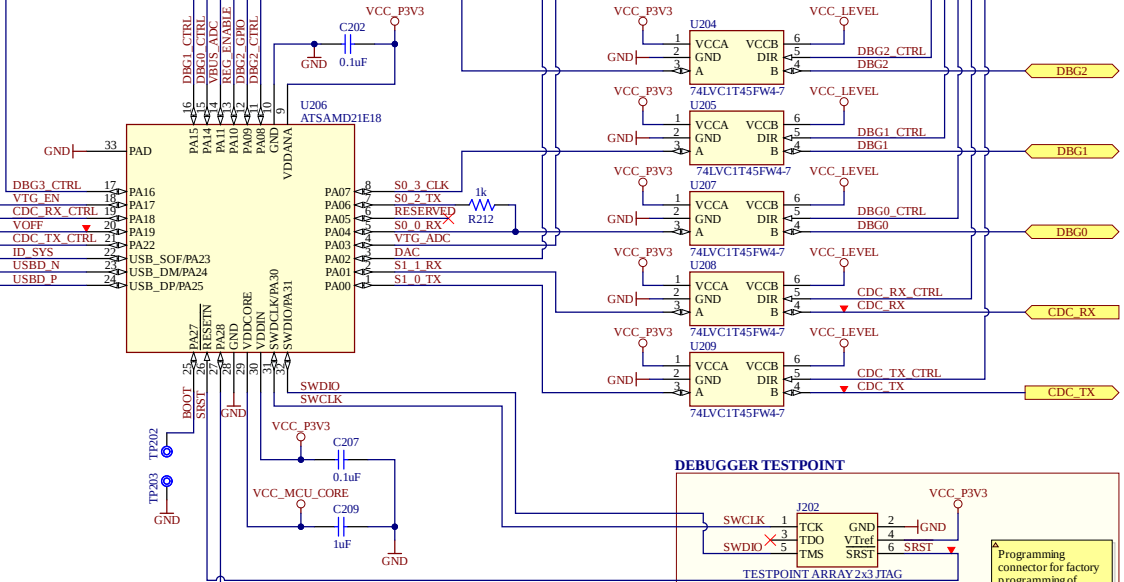
DEBUGGER USB-C CONNECTOR



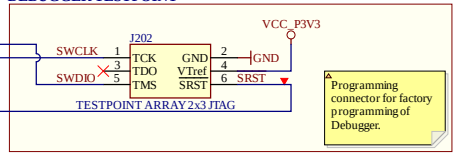
TP200
TP201
"GND"

TP200 and TP201 are MTG Holes in the corners of the PCB in the DEBUGGER section labelled "GND" on the silkscreen.

U206 ATSAMD21E18

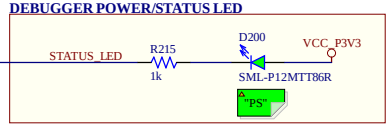


DEBUGGER TESTPOINT



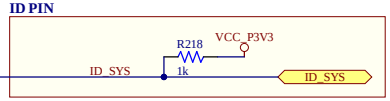
Programming connector for factory programming of Debugger.

DEBUGGER POWER/STATUS LED



TP200 and TP201 are MTG Holes in the corners of the PCB in the DEBUGGER section labelled "GND" on the silkscreen.

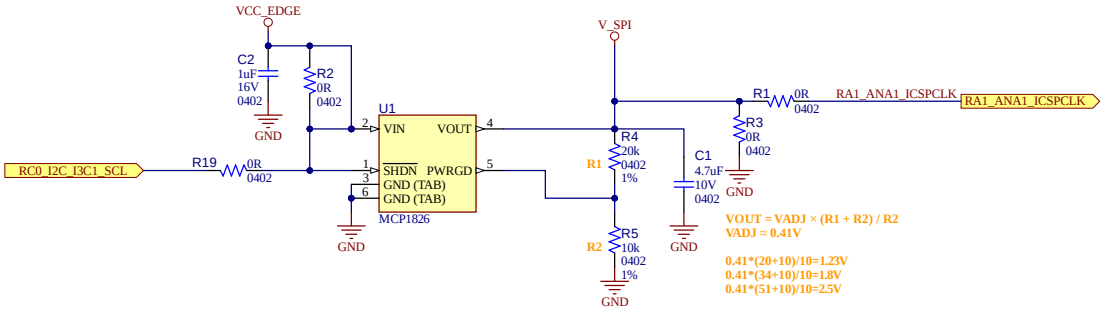
ID PIN



Project Owner: ST			
PCB Layout Contact: TF			
PartNumber: EV73T25A	Project Title: PIC18F16Q20 Curiosity Nano	Variant: [No Variations]	
Sheet Title: Debugger			
Size: A3	SCH #: 02-00457	Rev: 3	Date: 2026-10-30
	PCB #: 04-11563	Rev: 3	Sheet 3 of 5
File: PIC18F16Q20_Curiosity_Nano_Debugger.SchDoc			

CNANO Template Revision: 2.3

Designed with Altium
Altium.com



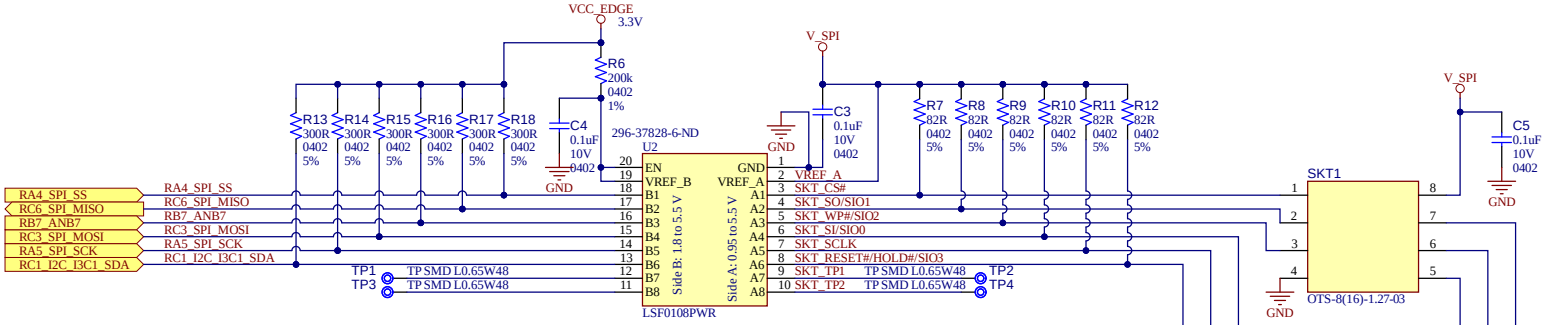


Table 9-3. Pull-Up Resistor Values

V_{PU} (1) (2)	8 mA		5 mA		3 mA	
	NOMINAL (Ω)	+10% (3) (Ω)	NOMINAL (Ω)	+10% (3) (Ω)	NOMINAL (Ω)	+10% (3) (Ω)
5 V	581	639	930	1023	1550	1705
3.3 V	369	406	590	649	983	1082
2.5 V	269	296	430	473	717	788
1.8 V	181	199	290	319	483	532
1.5 V	144	158	230	253	383	422
1.2 V	106	117	170	187	283	312

(1) Calculated for V_{OL} = 0.35 V
(2) Assumes output driver V_{OL} = 0.175 V at stated current
(3) +10% to compensate for V_{DD} range and resistor tolerance

Project Owner: ST				
PCB Layout Contact: TF				
PartNumber: EV73T25A	Project Title: *PIC18F16Q20 Curiosity Nano*	Variant: [No Variations]		
Sheet Title: **				Designed with Altium
Size: B	SCH #: 02-00457	Rev: 3	Date: 2026-10-30	Altium.com
	PCB #: 04-11563	Rev: 3	Sheet 4 of 5	
File: LevelTranslator_SOP_Socket.SchDoc				

Revision History

PCB Assembly Rev 1:

Design Changes:

Initial Design

PCB:

PCB revision 1

PCB Assembly Rev 2:

Design Changes:

Changed RA4 and RA5 net labels

PCB:

PCB revision 2
Silkscreen refers to the correct pins for SW0 and LED0.
Added UKCA logo to silkscreen

PCB Assembly Rev 3:

Design Changes:

- Changed USB connector to USB-C version
- Removed RC-filter and added start-up delay circuit instead
- Removed PTC and added current and slew-rate limter instead

PCB:

PCB revision 3
- Changed footprint for Level-shifters

CNANO Template Revision: 2.3

Project Owner: ST		MICROCHIP	
PCB Layout Contact: TF			
PartNumber: EV73T25A	Project Title: *PIC18F16Q20 Curiosity Nano*	Variant: [No Variations]	
Sheet Title: Revision History			Designed with Altium
Size: A3	SCH #: 02-00457 / PCB #: 04-11563	Rev: 3 / Rev: 3	Date: 2026-10-30 / Sheet 5 of 5
File: PIC18F16Q20_Curiosity_Nano_Revision_History.SchDoc			Altium.com

Comment	Description	Designator	Footprint	LibRef	Quantity
Capacitor NP		C1, C2, C3, C4, C5, C100, C101, C102, C103, C104, C105, C200, C201, C202, C203, C204, C206, C207, C208, C209	CAP_C_0402	CAP1684, CAP3233, CAP3201, CAP1033, CAP1208, CAP0616, CAP1265	20
Capacitor NP		C205	CAP_C_0603	CAP0543	1
LED_yellow	LED	D100	LED_0603_Yellow	DIODE1547	1
LED_green	LED	D200	LED_0402_GREEN	LED1092	1
Header 1x2		J107, J108	HDR_M_2.54_1x2	CON1632	2
CON USB2.0-C RCPT SMD RA KLS KLS1-5409		J203	CON_USB2.0-C_RCPT_SMD_RA_KLS_KLS1-5409	CON2580	1
08-10015		LABEL1	08-10015	08-10015	1
Transistor MOS-N GSD	Transistor MOS	Q200	X1-DFN1006-3	TRA1340	1
Transistor MOS-P GSD	Transistor MOS	Q201	LGA-3_GSD_0.7x0.6_CSD25501F3	TRA1611	1
Resistor	Resistor	R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R100, R101, R102, R103, R104, R105, R106, R200, R201, R202, R203, R204, R205, R206, R207, R208, R209, R210, R211, R212, R213, R214, R215, R216, R217, R218	RES_C_0402	RES2693, RSMT1238, RES2198, RES2904, RES2947, RSMT1273, RES2662, RES1022, RSMT1040, RES2661, RES1994, RSMT1163, RES1655	45
Socket SOIC-8	Socket SOIC-8	SKT1	SO-8_150mil_Socket	SKT1089	1
Switch 4 Pin	Switch	SW100	SW_TACT_SMD_KLS_L-KLS7-TS0304A-2-300-T	SWITCH1170	1
Mechanical		Test1, Test2		13-00211, 13-00018	2
Jupiter Test Jig		TESTFIXTURE1		15-00009	1
MCP1826T-FIXED SOT-223-5		U1	SOT-223-5	MIC9200	1
LSF0108PWR	Integrated Circuit	U2	SOP65P640X120-20N	LSF0108PWR	1
PIC18F16Q20 T-I/REB		U100	VQFN-20_3X3_EP1.8x1.8	MIC9477	1
MCP1711 UQFN-4	LDO MCP1711	U101	UQFN-4_1x1_EP0.48x0.48	MIC9478	1
MIC53XX ADJ MLF-6	LDO Regulator	U200	MLF-6 1.6X1.6mm	MIC7389	1
MIC94163_1.5MM_1MM		U201	MIC94163_1.5MM_1MM	MIC8744	1
MIC2008YML-TR VDFN-6		U202	VDFN-6_2x2_EP0.8X1.40	MIC9604	1
MIC5528MT DFN-6		U203	DFN-6_1.2x1.2	MIC6905	1
74LVC(H)1T45 SOT-23-6_SC-70-6_SOT-6_SOT-363_DSBGA		U204, U205, U207, U208, U209	X2-DFN1010-6	IC1665	5
SAMD21E QFN-32		U206	QFN-32_5X5_EP3.6x3.6_L	MIC8641	1